Step:4 Repeat the above steps for the reduced transportation table until all the allocations are made.

Note

If the minimum cost is not unique then the choice can be made arbitrarily.

Example 10.3

Obtain an initial basic feasible solution to the following transportation problem using least cost method.

	D_1	D_2	D_3	D_4	Supply
O_1	1	2	3	4	6
O_2	4	3	2	5	8
O_3	5	2	2	1	10
Demand	4	6	8	6	

Here O_i and D_j denote i^{th} origin and j^{th} destination respectively.

Solution:

Total Supply = Total Demand = 24

∴ The given problem is a balanced transportation problem.

Hence there exists a feasible solution to the given problem.

Given Transportation Problem is:

	D_1	D_2	D_3	D_4	Supply (a_i)
O_1	1	2	3	4	6
O_2	4	3	2	5	8
O_3	5	2	2	1	10
Demand (b_j)	4	6	8	6	

The least cost is 1 corresponds to the cells (O_1, D_1) and (O_3, D_4)

Take the Cell (O_1, D_1) arbitrarily.

Allocate min (6,4) = 4 units to this cell.

	D_1	D_2	D_3	D_4	a_i
O_1	(4) 1	2	3	4	6/2
O_2	4	3	2	5	8
O_3	5	2	2	1	10
b_j	4/0	6	8	6	

The reduced table is

	D_2	D_3	D_4	a_i
O_1	2	3	4	2
O_2	3	2	5	8
O_3	2	2	1	10
b_j	6	8	6	

The least cost corresponds to the cell (O_3, D_4). Allocate min (10,6) = 6 units to this cell.

	D_2	D_3	D_4	a_i
O_1	2	3	4	2
O_2	3	2	5	8
O_3	2	2	(6) 1	10/4
b_j	6	8	6/0	

The reduced table is

	D_2	D_3	a_i
O_1	2	3	2
O_2	3	2	8
O_3	2	2	4
b_j	6	8	

The least costis 2 corresponds to the cells (O_1, D_2), (O_2, D_3), (O_3, D_2), (O_3, D_3)

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